



OPEN ACCESS

EDITED BY

Ciro Rosolem,
São Paulo State University, Brazil

REVIEWED BY

Anderson De Souza Gallo,
Federal University of São Carlos, Brazil
Fasikaw Belay Mihretu,
Bahir Dar University, Ethiopia

*CORRESPONDENCE

Rathod Sridhar
✉ sridharrathod27@gmail.com

RECEIVED 15 November 2025

REVISED 08 December 2025

ACCEPTED 22 December 2025

PUBLISHED 16 January 2026

CITATION

Sridhar R, Longkumer LT, Pilla A, Bharteey PK, Jatav HS, Hareesh D, Vilakar K, Singh AP, Srikar K, Aruna K and Paarshitha Reddy MS (2026) Diverse fields for stronger yields: crop diversification strategies for sustainable agriculture and climate-resilient ecosystems. *Front. Agron.* 7:1746895. doi: 10.3389/fagro.2025.1746895

COPYRIGHT

© 2026 Sridhar, Longkumer, Pilla, Bharteey, Jatav, Hareesh, Vilakar, Singh, Srikar, Aruna and Paarshitha Reddy. This is an open-access article distributed under the terms of the [Creative Commons Attribution License \(CC BY\)](#). The use, distribution or reproduction in other forums is permitted, provided the original author(s) and the copyright owner(s) are credited and that the original publication in this journal is cited, in accordance with accepted academic practice. No use, distribution or reproduction is permitted which does not comply with these terms.

Diverse fields for stronger yields: crop diversification strategies for sustainable agriculture and climate-resilient ecosystems

Rathod Sridhar ^{1*}, L. T. Longkumer¹, Avinash Pilla², Prem Kumar Bharteey³, Hanuman Singh Jatav⁴, Dhara Hareesh⁵, Kayitha Vilakar⁶, A. P. Singh⁷, Katiki Srikar⁵, Kodavath Aruna⁸ and M. S. Paarshitha Reddy¹

¹Department of Agronomy, School of Agricultural Sciences, Nagaland University, Medziphema, Nagaland, India, ²Department of Agriculture, Koneru Lakshmaiah Education Foundation, Vaddeswaram, Andhra Pradesh, India, ³Department of Agriculture Chemistry, Chaudhary Chhotu Ram (Post Graduate) College, Muzaffarnagar, Uttar Pradesh, India, ⁴Department of Soil Science and Agricultural Chemistry, Sri Karan Narendra Agriculture University, Jobner, Rajasthan, India, ⁵College of Post Graduate Studies in Agriculture Sciences, Central Agricultural University (Imphal), Umiam, Meghalaya, India, ⁶Department of Soil Science and Agricultural Chemistry, Babu Jagjivan Ram Agricultural College, Professor Jayashankar Telangana Agricultural University, Telangana, India, ⁷Department of Agriculture, School of Agriculture and Development, Central University of South Bihar, Gaya, Bihar, India, ⁸Department of Agronomy, Professor Jayashankar Telangana Agricultural University, Rajendranagar, Hyderabad, India

In the context of altered climate regimes and escalating costs of cultivation, the conventional and non-cultivation practices have become economically untenable and unsustainable. This variability/fluctuations highlighting the need of adapting crop diversification strategies to promote sustainable agriculture practices and to maintain climate-resilient agroecosystems. Crop diversification helps to mitigate climate change impacts and supports the development of resilient and stable farming systems. Its underlying principles being systematic crop selection, resource conservation, optimal resource utilization, complementary crop combinations, own flourishing of the year-round planning of crop with various species of resources of surplus, not compromising yield and optimizing yield in resource-deprived drylands and rainfed areas. A systematic review of 134 diversified systems, obtained using a PRISMA guided meta synthesis from 2010 to 2025, shows that such systems produce average yields that are 20-38% better than monocropping and builds soil organic carbon by 9% with a reduction of 25-40% synthetic inputs. These study reveals that diversified systems viz., intercropping and agroforestry reliably boost soil health, increase biodiversity, reduces dependence on chemical inputs and consequently improve climate adaptation capacity along with socio-economic conditions. Crop diversification generally reduces the incidence of pests and disease due to increased population of natural enemies which disrupts the pest activity compared to monoculturing which makes more susceptible to pests and diseases. Nevertheless, the effectiveness of the diversification is moderated by regional climatic conditions, policy frameworks and access to markets. Adoption is further hindered by knowledge deficit, infrastructural limitations and lack of risk aversion strategies amongst smallholders. This

review addresses these gaps by offering a systematic global assessment of the benefits of diversification and the constraints to adoption, highlighting that the scaling of diversification processes requires context-specific policy incentives, knowledge transfer to farmers and value chain development for non-traditional crops.

KEYWORDS

biodiversity, ecosystem, mitigation, PRISMA, resource optimization

1 Introduction

Climate change represents one of the major challenges to global food security by modifying agricultural systems through a variety of mechanisms, such as rising mean temperatures, unpredictable monsoon, precipitation variability and enhanced weather extremes, which creates environmental stress to crop productivity (Erickson et al., 2009; Gomez-Zavaglia et al., 2020; Misra, 2014). Climate change is a major focal point for declining crop productivity, nutrient depletion, pest and disease infestations due to deterioration of the natural resources. Consequently, modern intensive monoculture and chemical farming have become very susceptible to a wide range of biotic and abiotic stresses leading to severe yield loss. Crop diversification is the planned strategy and management of alternative crops/cropping systems in an agricultural production system on a farm in order to achieve the highest sustainable benefits (Clements et al., 2011). It acts as a building block of sustainable intensification through restoration of biotic interactions, mitigation of yield-emission trade-off and adoption of ecologically complementary species into cropping systems (Nandi et al., 2024; Dowling et al., 2021). Similarly, it offers ecosystem services (Tamburini et al., 2020; Beillouin et al., 2021).

1.1 Crop diversification strengthens ecosystem resilience to climate change and biodiversity loss

Crop diversification is an alternate solution in terms of economic returns while maintaining soil health and sustaining the livelihood of farmers (McCord et al., 2015). Crop diversification is now regarded as a game-changer, offering an ecological and sustainable alternative to the high-risk resource-intensive monoculture practice (Beillouin et al., 2019; Mortensen and Smith, 2020). For instance, legume intercropping systems have been proven to raise significant yields, decrease pest and disease levels by breaking pest life cycles, enhance soils while decreasing input (Shah et al., 2021a; Chamkhi et al., 2022; Zou et al., 2024). Restoration of natural habitats and limitation of synthetic input use to develop climate smart food production (Lipper et al., 2014;

Muhie, 2022). For example, maize alongwith legumes such as lupine or cowpea in strip cropping system, enhances biological nitrogen fixation and reduces pest-disease outbreaks (Dowling et al., 2021 and Maitra et al., 2021, respectively). Results of the study recommends to select complementary based crop associations that favours the nutrient use efficiency (NUE), crop growth dynamics, resilience against native pest and diseases, ensuring crop diversity and ecological balance (Beillouin et al., 2021; Cozim-Melges et al., 2024; Grahmann et al., 2024; Gawdiya et al., 2025). Such diversification methods, according to agroecological principles, help in the sustainability of the ecosystem, development of governance structures and resiliency of supply chains (Vernooy, 2022). Diversification of crops has emerged as a key strategy for achieving the sustainability and resilience of agricultural systems, as shown in Figure 1.

1.2 Monocropping under climate stress and the adaptive necessity for diversifying crops

Contemporary monoculture systems have demonstrably deteriorated the crop yields, soil productivity, ecosystem diversity and functional integrity (Bybee-Finley and Ryan, 2018; Pretty and Bharucha, 2014). In the face of growing climate vulnerability, the need and urgency for the use of adaptive management strategies to adopt crop diversification, intercropping, crop rotation, agroforestry, cover cropping, conservation tillage and water-sparing irrigation to meet the limitations of conventional monoculture, which help to buffer climatic disturbs through increased resource-use efficiency, moderation of microclimates and yield stabilization.

1.2.1 Temperature changes

Heat waves in the atmosphere are absorbed by infrared-active gases, mainly carbon dioxide (CO₂), ozone (O₃), and water vapor (H₂O), which subsequently warm up the Earth in a phenomenon recognized as the greenhouse effect (Malhi et al., 2021). Since 1850, the average global temperature has increased by 1.1 to 2 °C which posed a serious threat to sustainability of agricultural systems and food availability. Increasing temperatures have impacted agricultural productivity through multiple processes (Elahi et al.,

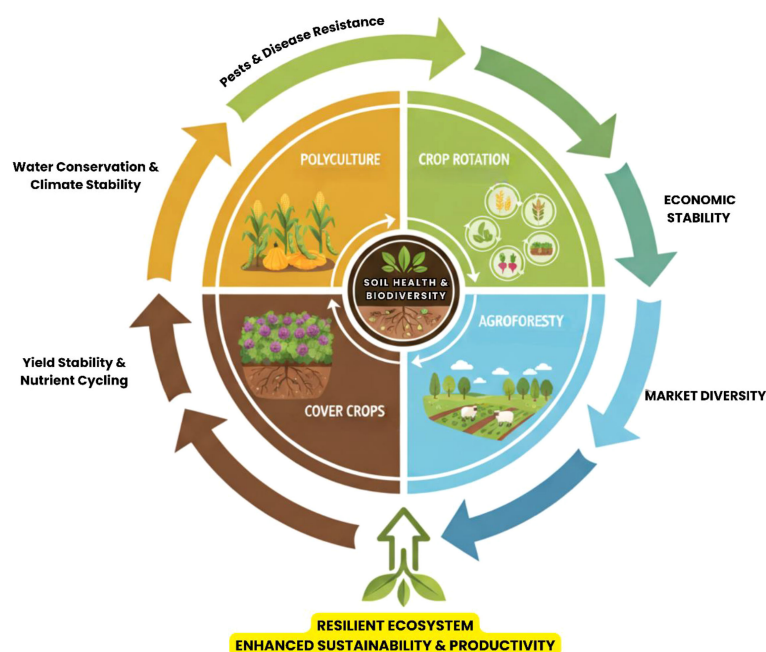


FIGURE 1
Diversified cropping system for a resilient ecosystem.

2022), including increased water stress, changed crop phenology, and increased pest and disease pressure, reduce crop yields, degrade crop quality and interferes with pollination at crucial growth phases (Nawaz et al., 2024a). Furthermore, hot weather exacerbates vulnerability of heat sensitive plants like wheat, rice, and maize (Ahmed et al., 2019). On the other hand, severe cold occurrences, especially in Pakistan's north, have also endangered agricultural systems because frost and freezing temperatures deteriorates plant tissues, reducing yields and quality and quantity of crops (Hassan et al., 2021).

1.2.2 Precipitation changes

Significant shifts in precipitation patterns have the potential to harm infrastructure and reduce agricultural productivity. Unusual rainfall has harmed the ripe crops, while droughts cause a decline in agricultural productivity and food security in many areas (Saleem et al., 2024). A recent study in Ethiopia reported that decreased maize and teff yields resulted from increased rainfall variability (Temesgen et al., 2021). Similarly, reduced rainfall in Sub Saharan African region led to lesser maize productivity, decreased precipitation has resulted in a reduction in maize crop yields, which is the main staple food in the region (Chapman et al., 2020). Intense rainfall events and flooding has led to soil erosion, nutrient leaching and water logging, all of which can hinders crop health and reduced yields (You et al., 2024).

1.2.3 Long-term repercussions of adverse climate shifts

Disasters associated with changing climate have the potential to destroy crucial infrastructure, significant public assets and crops,

which would be harmful to both domestic revenue and food security. The amount and consistency of irrigation water available, as well as the unpredictable pattern of floods and droughts, will all be impacted by the rapid thawing of glaciers (Davidson, 2018). The aquatic food web is impacted by changes in the nitrogen cycle, plankton productivity and ocean warming (Azani et al., 2021). The greatest impact is recorded in low-income nations and regions that are already prone to food insecurity; this results in food shortages, a decline in the nutritional quality of food and long-term negative health effects (Nawaz et al., 2024b).

1.2.4 Recurring disease outbreak (Frequent disease outbreak)

It is estimated that an increase of one degree temperature will leads to a 10–25% rise in losses due to insect pest invasion (Shrestha, 2019). Changing weather has the potential to boost pest numbers and relocation habitat, which could have deleterious effects on agricultural viability and production, as the pest population is mostly reliant on abiotic variables like temperature and humidity (Bradshaw et al., 2024). As air humidity increases, the fungus *Sclerotinia sclerotiorum* becomes more pathogenic; disease growth in lettuce plants peaks when air relative humidity reaches 80% (Szyniszewska et al., 2024). Fluctuating temperatures have a greater impact on several forest diseases (Sturrock et al., 2011). Climate change has made more places conducive to pest invasion (Saleem et al., 2024). The habitat appropriateness of the three common African bug species, *Tuta absoluta*, *Ceratitis cosyra* and *Bactrocera invadens*, is rising across the entire continent, particularly in regions near their ideal habitat (Biber-Freudenberger et al., 2016). Climate

change also affects the problem of increased agricultural weed infestation. C_3 weeds respond more violently as the CO_2 content rises (Saleem et al., 2024). The impact of climate change has a deleterious effect on the weed population as the expansion of geographic ranges is recorded. The management will only be possible through development of innovative strategies that explicitly account for these varied climatic conditions (Malhi et al., 2021).

1.3 Crop diversification and ecosystem resilience

Climate change has caused extreme temperatures, frequent and intensive floods, cyclones, and other natural disasters, which are expected to worsen. Crop diversification can protect natural biodiversity, strengthen the agro-ecosystem's ability to respond to these stresses, minimise environmental pollution, reduce the risk of total crop failure (Lakhran et al., 2017).

1.4 Ecological benefits, economical benefits, and social benefits

Integration of community practices with new and advancing technologies, these systems promote a pathway towards sustainable resource management and livelihood generation (Akther and Evans, 2024; Koontz et al., 2015). Ecological issues such as declining biodiversity, disruption of natural species relationship, change in crop ecosystems, increased greenhouse gases emission from monocropped ecosystems are some of the issues which highlight the need of finding out novel practices and improved management strategies which shall address such challenges in a new socio- economical way. Crop diversification is able to diminish these challenges and thus acts as a strong alternative to monocropping. Economically, diversified cropping helps to alleviate crop failure and therefore mitigate financial risk for farmers while net returns at the farm level can rise by 15–25% and chemical inputs can be decreased by 30%, contributing to employment generation driven by labour-intensive operations on the farm (Adam and Abdulai, 2024). Socially it supports gender inclusiveness and local food systems stability.

Traditional knowledge, collaborative decision-making and policy reforms go a long way in strengthening resilience, especially when institutional inertia is the challenge when it comes to progress (Shammin et al., 2022). Grassroots innovation, often, is the force behind localized solutions where citizen-led work and social enterprises go hand in hand to bridge gaps in environmental conservation and economic inclusivity (Royse et al., 2024). Harnessing local knowledge and collaborative efforts is not only a way of solving immediate ecological challenges but can also set up sustainable opportunities for entire communities (Gawdiya et al., 2025). Nonetheless, premium incentives are highly recommended for the government towards stakeholders

instead of focusing solely on subsidies (Gawdiya et al., 2025). A changing environment also makes rural people more vulnerable since communities have few alternative ways of life and small family farms cannot afford costly adaptation schemes. Researchers need to develop resilient agricultural systems through reasonable and inexpensive approaches to maintain ecosystem functions and services, along with livelihoods (Lakhran et al., 2017).

1.5 Challenges that are hindering the adoption

The monocultural dominance is driven by interconnected social-ecological systems, cultural norms and economic dynamics (Pretty, 2011; Song et al., 2021; Souissi et al., 2024). Various factors set the tone for the production of commodity crops, which is conducive to the increase of wealth, the strengthening of economic inequalities and the development of agricultural specialisation through tightly integrated supply chains and a policy support (Baffes and Nagle, 2022; Briones and Rakotoarisoa, 2013; Kastner et al., 2021). Corporate driven research that favors input dependent farming has been perpetuating a cycle of Path-dependency that hinders farmers ability to diversify and ignores systemic constraints that farmers face (Aare et al., 2021; Whitton and Carmichael, 2024). Most research on diversification adoption is individual level based on specific aspects, while overlooking the systemic impacts (Bogado et al., 2024). Depending on the circumstances, these factors have disparate effects and are less likely to predict a behavior in adoption models (Bernzen et al., 2023). This has led to an increased awareness of interconnected institutional and structural barriers hindering the diversification (Gawdiya et al., 2025). Continuous monoculture practices often lead to landscape degradation, degrades landscapes, while sustainable intensification through diversification enhances yields over multiple growing seasons and provides significant climate benefits (Crews et al., 2018; Vernooy, 2022). Monocultures may be more common in the agricultural system because of the availability of and influence of extension agents. Reluctance to deal with various crops today is due to scarcity of scientific information about suitable agronomic techniques, such as climate-smart genotypes, environment-management interaction and its effects on ecosystem services, and poor governance (Gawdiya et al., 2025). To quantify this analysis, Simpson Index of Crop Diversification (SICD) was used based on the FAO production area statistics (1990–2023) for 5 Indian regions (Kumar and Gupta, 2015). The higher SICD values (> 0.6) denote diversified patterns, lower values (< 0.3) specialization. Observed data showed gradual diversification in eastern India or plateauing in north-western plains dominated by rice-wheat systems as shown in Figure 2. The present review highlights the ecological and agronomic benefits of crop diversification as a whole sustainable agriculture. Researchers across the world have emphasised the importance of diversification in the restoration and maintenance of ecological balance in farming systems.

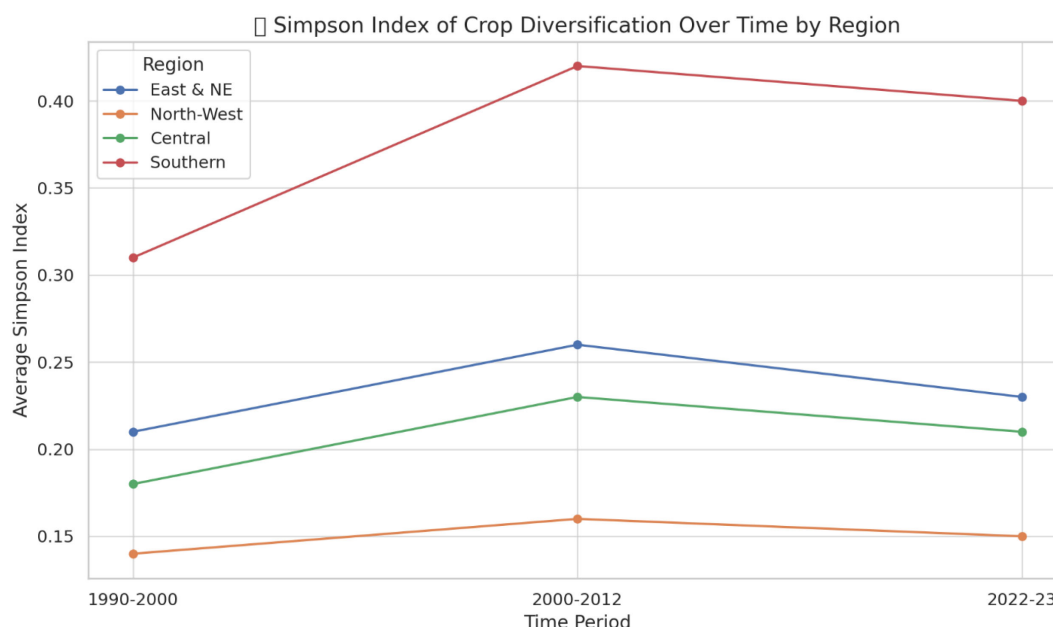


FIGURE 2

Simpson index of crop diversification over time by region (1990–2023); Source: Based on authors' calculations (Kumar and Gupta, 2015).

2 Methodology

To synthesize rigorously and comprehensively the role of crop diversification in boosting ecosystem resilience, a systematic review literature was carried out per the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. The search strategy was designed to cover as broad a range of peer-reviewed research as possible relevant to agroecological systems, climate resilience and diversification practices. Searches have been conducted in several major academic databases including Web of Science (Clarivate Analytics), Scopus (Elsevier), Science Direct, CAB Abstracts (via CABI), PubAg (USDA), Google Scholar, SpringerLink, Taylor and Francis Online and JSTOR. These databases were chosen because they have good coverage of environmental sciences, agricultural research and interdisciplinary studies on sustainability areas.

2.1 Concepts

A set of boolean search strings was created using combinations of primary and secondary keywords in order to ensure that the retrieval of relevant literature was exhaustive. Core search terms included “crop diversification”, “ecosystem resilience”, “agroecological systems”, “climate adaptation”, “intercropping”, “sustainable agriculture”, “crop rotation”, “soil health”, “biodiversity”, “carbon sequestration” and “resilience metrics”. These keywords were searched in different combinations in the databases and updated monotonically in order to maximize precision and recall. In order to ensure that the temporal scope is

appropriate as well as to capture recent advancements, only studies published between January 2010 and March 2025 were included.

The systematic search produced an initial corpus of 1,350 records. Of these, there were 1,240 identified by database searches and 110 identified using manual reference inspection and retrospective citation search. After removing duplicates ($n=170$), a total of 1180 unique articles were taken through the title and abstract screening process. According to their relevance to study objectives, 910 records were excluded. The remaining 270 papers were identified for full text review, along with 100 reports that were manually identified following an ancillary search process. After exclusion criteria (non-peer-reviewed materials, lack of an empirical focus, or lack of methodological detail) were applied, 134 studies met inclusion criteria and were included in qualitative synthesis. Selection of studies was guided by stated inclusion criteria: (i) peer-reviewed articles 2010–2025; (ii) English; (iii) empirical, modeling-based, or synthetic reviews directly relevant to crop diversity and ecosystem resilience; and (iv) geographical breadth, including agroecological zones semi-arid tropics, monsoon climate, temperate, and drylands. Key outcome measures included soil organic carbon (SOC), microbe biomass carbon (MBC), nutrient cycling indicators, crop yield stability, pest and disease resistance, pollination service and climate-stress tolerance. Studies that were solely monoculture based or whose full texts were not publicly available were excluded from the analysis. Data were extracted using a structured coding framework that included publication metadata (author, year, journal), geographical scope, agroecological zone, diversification typology (i.e. intercropping, rotation, agroforestry) and quantitative and qualitative outcome measures. Data were collated using Microsoft Excel and Zotero was used for reference management. Theme synthesis of extracted data

revealed patterns in ecological, economic and social dimensions whereas meta-summarization methodology revealed a frequency-weighted output of crop-diversification policy outcomes. Hence the final set of 134 studies has become a basis of a comprehensive thematic analysis that allowed for critical evaluation of the role of crop diversification in ecosystem functions related to soil fertility, biodiversity conservation and climate mitigation. A PRISMA flow diagram is shown in Figure 3 to illustrate the review process and shows the stages of identification, screening, eligibility assessment and inclusion. “Quotation marks in the list of keywords were used solely for highlighting purposes and were not part of the database search syntax.”

2.2 Data extraction and synthesis

Following a full text screening phase, the chosen studies were subjected to a well defined data extraction process, using a pre-specified coding scheme that was developed to ensure cohesion, transparency and reproducibility for the extraction of important study features and outcomes. Each publication included was systematically analysed to extract specific metadata and

contextual information including author(s), year of publication, geographical focus (e.g. India, Sub-Saharan Africa, Europe), agroecological zone (drylands, humid tropics, temperate regions) in which study was located. The type of diversification practice used (i.e., mixture practices and cropping systems) was noted, as was the type of crops used such as mixed cropping systems such as maize-legume, millet-pulse and rice-wheat systems and rotation and agroforestry. As indicators of environmental and agronomic performance, a number of soil health parameters were measured including soil organic C (SOC), microbial biomass C (MBC), and dehydrogenase activity, which collectively provide an indicator for dynamics of biological activity and nutrient turnover in the soil. Significance of yield was expressed as percentage gains or losses with regard to conventional monoculture baselines. Additionally, resilience-related metrics including yield stability under stressors related to climatic influences and occurrence of pest and disease outbreaks were reported in order to capture the adaptive benefits of diversified systems. The extracted data were further organised by Microsoft Excel, for tabulating and cross-comparison and Zotero for citation management and referencing. Qualitative thematic analysis and meta-synthesis were used to extract emergent patterns and cohesive conclusions among studies. This allowed

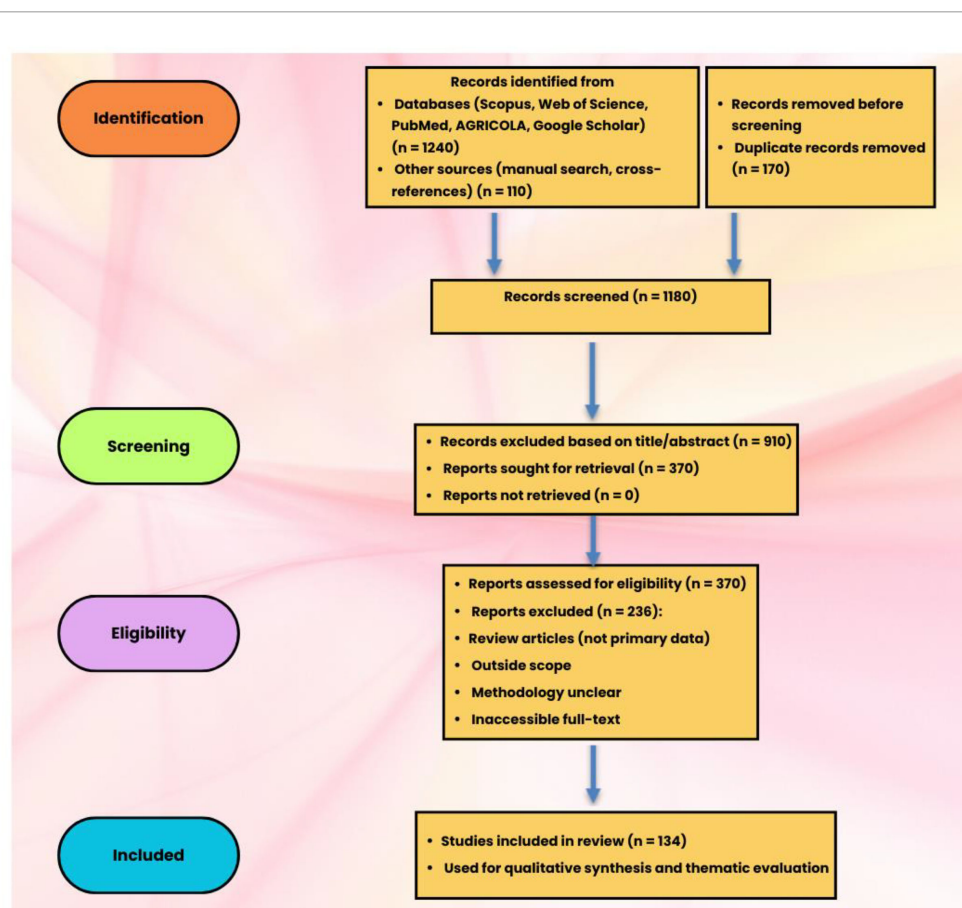


FIGURE 3

A PRISMA flow diagram was employed to detail the systematic protocol followed for study selection in this review. (2010–2025 PRISMA set; n=134 studies).

for a multidimensional gauge of the potential benefits of crop diversity towards ecological, agronomic and resilience outcomes across different agroecosystems.

3 Thematic synthesis

3.1 Ecological benefits

3.1.1 Biodiversity enhancement:

Diversified cropping systems promote a wider range of flora, fauna and beneficial insect species, soil microorganisms, birds and pollinators (Altieri, 1999). The area's biodiversity is a source of key ecosystem services, including pollination, natural pest control and nutrient cycling. Intercropping and agroforestry have been empirically shown to be effective strategies for increasing on-farm biodiversity (Kremen and Miles, 2012). Diversified cropping systems can have different efficiency in their effectiveness, with

agroforestry the most efficient in providing multiple ecosystem services. Diversification of cropping pattern has been found to increase biodiversity by 24% and promote ecosystem services, including improvement of water quality (84%), a decrease of pest-disease incidence (63%) and a better soil quality (11%) (Beillouin et al., 2021). Besides, a diversified crop system enhances soil organic carbon and microbial biomass carbon by 9% and 12% compared with monoculture (Yan et al., 2023). Nitrogen (N)-fixing rotations have decreased synthetic N inputs through 25-40%, whereas agroforestry sequestration was 1.2 Mgt CO₂ e ha⁻¹ yr⁻¹ more than single crops (Cardinael et al., 2022). Crop diversification practices viz., crop rotation, multiple cropping, intercropping and integration of grain legumes in cereal-based production systems to increase productivity, stability and providing ecosystem services while also promoting sustainable production systems (Nasiro, 2024). These agricultural methods are increasing agro-biodiversity by stabilizing species richness and ecological interactions in farming systems. As Figure 4 shows, the diverse cropping systems and agroforestry

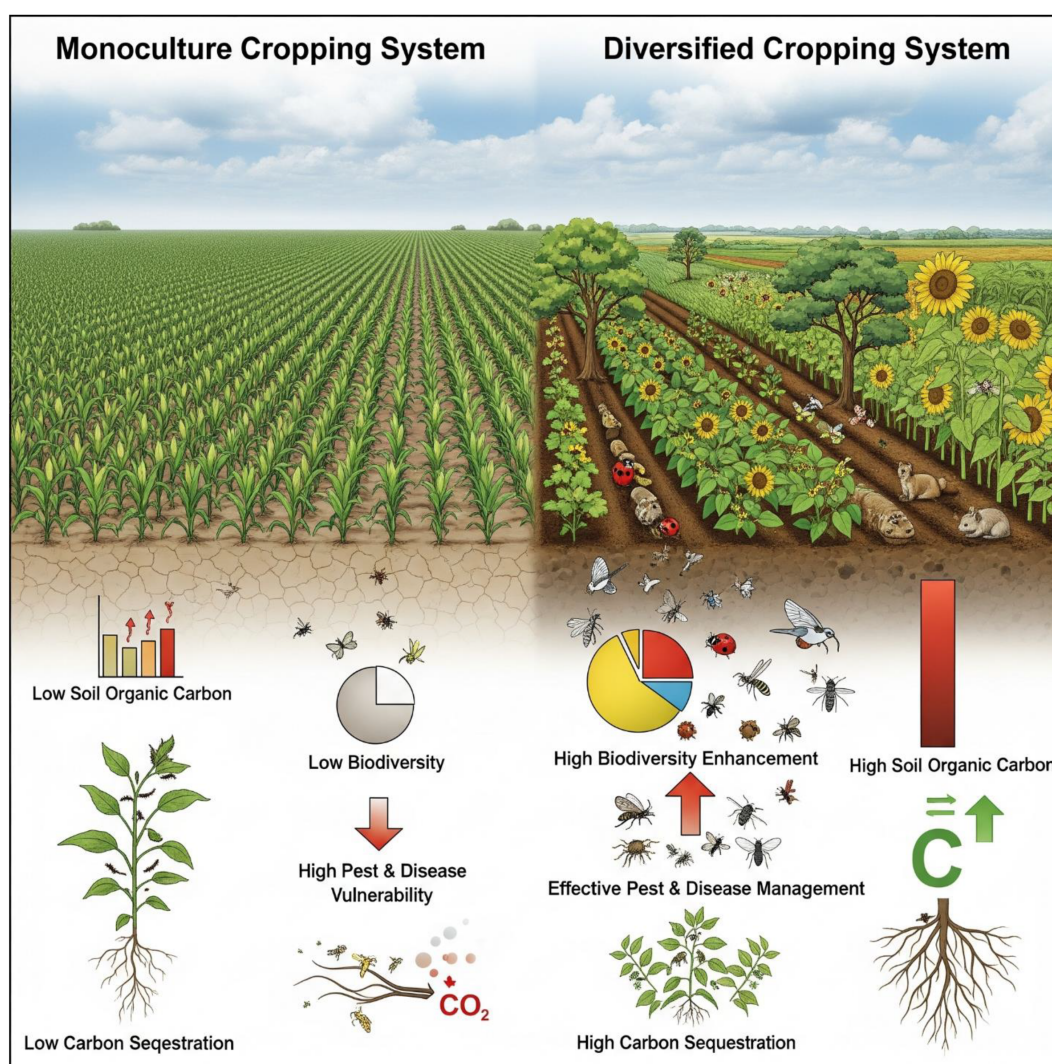


FIGURE 4
Monoculture cropping systems vs diversified cropping systems, their roles in the ecosystem and on soil.

promoted beneficial organisms including pollinators, natural enemies of pests and organisms living in soil. According to a recent analysis by [Tamburini et al. \(2020\)](#) and [Sridhar et al. \(2025\)](#), mixed cropping systems have shown to promote much more biodiversity than monocultures.

3.1.2 Soil health improvement:

Crop diversity of farm rotation improves soil structure, organic matter content and microbial diversity. Legumes included in crop rotation fixes atmospheric nitrogen and therefore reduces the use of synthetic fertilizers ([Drinkwater et al., 1998](#)). Root systems of diverse crops reach various depths and they enhance soil aeration and availability of nutrients. Such diversity also breaks pest and pathogen cycles, which helps relieve pressure from disease. Consequently, enhancement of soil structure, microbial activities and organic matter augmentation following crop diversification, so increase long term production. Integrating legumes with other crops in the sequence enhances nitrogen availability in soils and carbon reserves, reducing the need for chemical fertilizers ([Beillouin et al., 2021](#)). Diversified farming practice has been found to increase carbon accumulation in agroecosystems compared with monocultured farming. The content of macroaggregates and the carbon content increased by 5.0 and 12.5% of the contents, respectively, while the contents of microaggregates and silt/clay fractions decreased under diversified cropping ([Li et al., 2023](#)). The implementation of diversification of cropping systems, especially with legumes and oilseeds, in association with conservation tillage, not only improved soil health and productivity, but the carbon footprint of these sustainable practices for Zero Tillage (ZT) and Permanent Bed systems (PB) was 465% and 822% respectively, higher than for Conventional Tillage (CT) demonstrating the environmental benefits of these practices in calcareous soils ([Pramanick et al., 2023](#)). Carbon sequestration, mainly attributed to the carbon of microbial biomass, improved, whereas soil quality improvement was mostly attributed to soil organic carbon and aggregate stability. These findings argue that diversified cropping may have positive consequences for agroecosystems because it increases soil health and carbon storage at the same time ([Yan et al., 2023](#)). The incorporation of wide variety of crops including legumes, oilseeds, etc., increases the soil health by altering physical, chemical and biological properties. Moreover, different root architectures of various crops increase the soil porosity and nutrient distribution in soil. These improvements not only contribute to improving soil fertility and productivity but also enable the soil to store carbon more effectively, thereby contributing to climate mitigation.

3.1.3 Water management:

Crop diversification results in efficient water utilization by optimizing root structures and water transpiration patterns. Deep-rooted crops reach water resources of lower soil horizons, and shallow-rooted crop species help retain soil moisture. Cover crops decrease runoff and enhance infiltration and it promotes better groundwater recharge ([Daryanto et al., 2016](#)). Construction of small-scale water-harvesting infrastructure and management

practices to maximize soil water availability and water use by crops, addresses challenges that come with climate change and water resources pressures ([Molénat et al., 2023](#)). Incorporation of spring crops resulted in a 7-12% decrease in annual actual crop ET and a 21-31% decrease in net use of ground water relative to the traditional winter-wheat-summer-maize double cropping system ([Wang et al., 2024](#)). These practices offer long-term benefits, including the enhancement of soil microbiological activity, reduced fertilizer application, maintenance of low water consumption and crop quality stability ([Lóczy et al., 2024](#)). Crop diversity benefits other people in selecting the right agronomic systems in a particular region and also helps with water management, with more water retained in the soil and less lost by evapotranspiration. Cumulative evidence validates the idea that crop diversification leads to more efficient use of water and helps build ecological resilience, and thus is a vital element of climate-smart and sustainable agricultural systems.

3.1.4 Carbon sequestration

Crop diversification for improving soil carbon inputs through plant residues and root biomass. Practices such as agroforestry, cover cropping and conservation tillage add to the soil organic carbon (SOC) pools ([Lal, 2004](#)). Trees and perennials in diversified systems are long term carbon sinks, contributing to the mitigation of climate change. Intercropping of rainfed woody crops with perennials enhances SOC and nitrogen (N) storage and stabilization, suggesting a positive approach to promote soil fertility and to mitigate the impacts of climate change in semi-arid climates ([Almagro et al., 2023](#)). Diversified cropping schemes store soil carbon up to 1.2 Mg C ha⁻¹ y⁻¹ more than monoculture. Incorporation of agroforestry, perennial cropping and organic amendments leads to an increase of above- and below-ground carbon stores, making diversification a climate smart strategy ([Cardinael et al., 2022](#)). These systems enhance SOC stock by 9% while enhancing a number of topsoil biochemical indicators, but require 40% less fertilizer than conventional wheat/maize systems. The improvement is mostly attributed to an increase in SOC content of large macroaggregates and enhanced microscopy turnover due to the variety of inputs from fresh residues ([Yan et al., 2023](#)). Agroforestry encourages carbon sequestration through stable deep root systems and biomass storage, while crops diversification and rotation further the cycling of nutrients and maintain a diverse collection of soil microorganisms; however, the effectiveness of all these processes can differ in different environments ([McCauley and Barlow, 2023](#)). The carbon sequestration is highly affected by soil management. Zero tillage, retention of crop residue, crop rotation, application of organic manures and biochar could be used to increase soil organic carbon formations and decrease the CO₂ emissions. These actions enhance the soil health and promote the long-term and climate-resilient carbon storage ([Dasgupta and Mahanty, 2024](#)).

3.1.5 Mitigation of greenhouse gas emissions

Diversified systems reduce the need for synthetic fertilisers and pesticides, and thus reduce the emissions of nitrous oxide (N₂O)

and methane (CH₄). By improving nutrient cycling and decreasing tillage further greenhouse gas emissions are reduced. Crop diversification with legumes can considerably reduce the carbon footprint of agriculture (Tilman et al., 2002). Reducing fertilizer and pesticide inputs, boosting nitrogen use efficiency and limiting tillage frequency are all ways in which diversified systems can cut agricultural greenhouse gas emissions by up to 30%). The integration of legumes and conservation tillage are especially effective strategies (Smith et al., 2020; Kanter et al., 2021). Integrating cash crops and legumes in crop rotations with traditional cereals increased equivalent production by up to 38% and cut N₂O emissions by 39% and improved the overall greenhouse gas balance of the system by 88% (Yang et al., 2024; Jiang et al., 2022). Agriculture is the second most important sector in terms of greenhouse gas emission, after the energy sector, with animal production, fertilisers high in N, combustion of crop residues and water management being important sources. Implementing greenhouse gas mitigation strategies in agriculture is fundamental to curb its contribution to global emissions, which could lead to a significantly higher benefit both for the environment and in food production (Kabange et al., 2023). The agricultural sector accounts for 13 per cent of the global anthropogenic greenhouse gas emissions (mostly due to animal production, rice cultivation and the use of synthetic fertilisers) which underlines the need for effective mitigation techniques. Research shows that through improved nutrient management, organic agriculture, conservation tillage, and carbon sequestration, the rate of greenhouse gas emission curtailments can be reduced by up to 89% with enhanced SOC (M et al., 2024).

3.1.6 Pest and disease management:

Monocultures are the result of homogenous habitats that promote pest and diseases outbreaks. On the contrary, pest life cycles are broken and habitats for natural predators and parasitoids are created in diversified cropping systems (Letourneau et al., 2011). Crop rotation and intercropping reduce the risk of the development of resistance in pests and maintain the ecological balance (Jaworski et al., 2023). Agro-ecological strategies like crop diversification and habitat modification reduce pest & disease risk by breaking pest life cycles & easing natural enemies & increase biodiversity & reduce reliance on chemical pesticides (Chellam et al., 2024). Introducing flexible cropping patterns reduces the various risks associated to pests and diseases (Lenné and Wood, 2024). Crop rotation has a big effect on the soil microbiome and enhances the community of the bacteria, which suppresses soil-borne diseases. Variation in plant inputs to soil organic matter pools may be a driver of changing patterns of microbial communities and an enhancement in disease suppressive functional potential in response to crop diversity (Peralta et al., 2018). Diversified cropping systems have lower pest and disease incidence compared to monoculture due to the higher population of natural enemies interfering with pest and disease life cycles.

3.2 Cultural and socio-economic benefits

3.2.1 Economic resilience and stability of income

Crop diversification is an empirically documented mechanism of farm income stabilisation and resilience to market fluctuations and climatic shocks (Hien, 2025; Staniszewski and Borychowski, 2020; Dagunga et al., 2023). By diversification, in particular, the cultivation of a diverse mix of crops enables farmers to diversify risk sources, hence producing more consistent household incomes (Zhang et al., 2024; Sain et al., 2023; Makate et al., 2016). Recent empirical studies in the North China Plain show that diversified crop rotations that include cash crops and legumes can increase yields by up to 38% and net farm income by almost 20%, while at the same time producing significant environmental benefits (Zhang et al., 2024; Dagunga et al., 2023). These benefits are of stronger relevance to smallholder and marginal farmers who are exposed to a higher level of risk and chronically benefit from diversification across species of crops and market channels (Lopez et al., 2024; Ndlovu et al., 2024). In many developing country settings, crop diversification is also regarded as a powerful remedy for poverty reduction, due to its ability to simultaneously raise incomes and nutritional status (Hien, 2025). The economic benefits offered by different forms of crop diversification as shown in Tables 1, 2 clearly indicate that diversified systems bring more economic benefits compared with monoculture arrangements.

3.2.2 Market access and value chains

Market access and value chains form determining factors in crop choice, hence important to be considered carefully in order to prevent marketing challenges linked with produce. Successful promotion of diversification of crops, especially at the grassroots level, requires a focus on improving post-harvest handling and storage facilities which leads to value addition along lines of supply chain and simultaneously spurs employment generation (Lopez et al., 2024; Shah et al., 2021b). Enhanced connectivity and connectivity open new avenues for markets for produce which equip farmers with the option of a wider range of crop choice combinations, alternative crops, while the creation of markets for non-traditional varieties allows them to transit from subsistence to commercial agriculture (Das et al., 2024; Negi et al., 2020). Collaborative efforts between government entities, private sector and farmer organisations are imperative to build robust value chains and also ensure fair returns from diversified crops (Sain et al., 2023; Dagunga et al., 2023). A complete knowledge of the existing market dynamics and strategic crop choice in diversification always provide a remunerative edge when compared to monocropping.

3.2.3 Food security and nutritional enhancement

Crop diversification has a direct impact on food security by increasing the availability and accessibility of a range of different

TABLE 1 Types of crop diversification and their benefits.

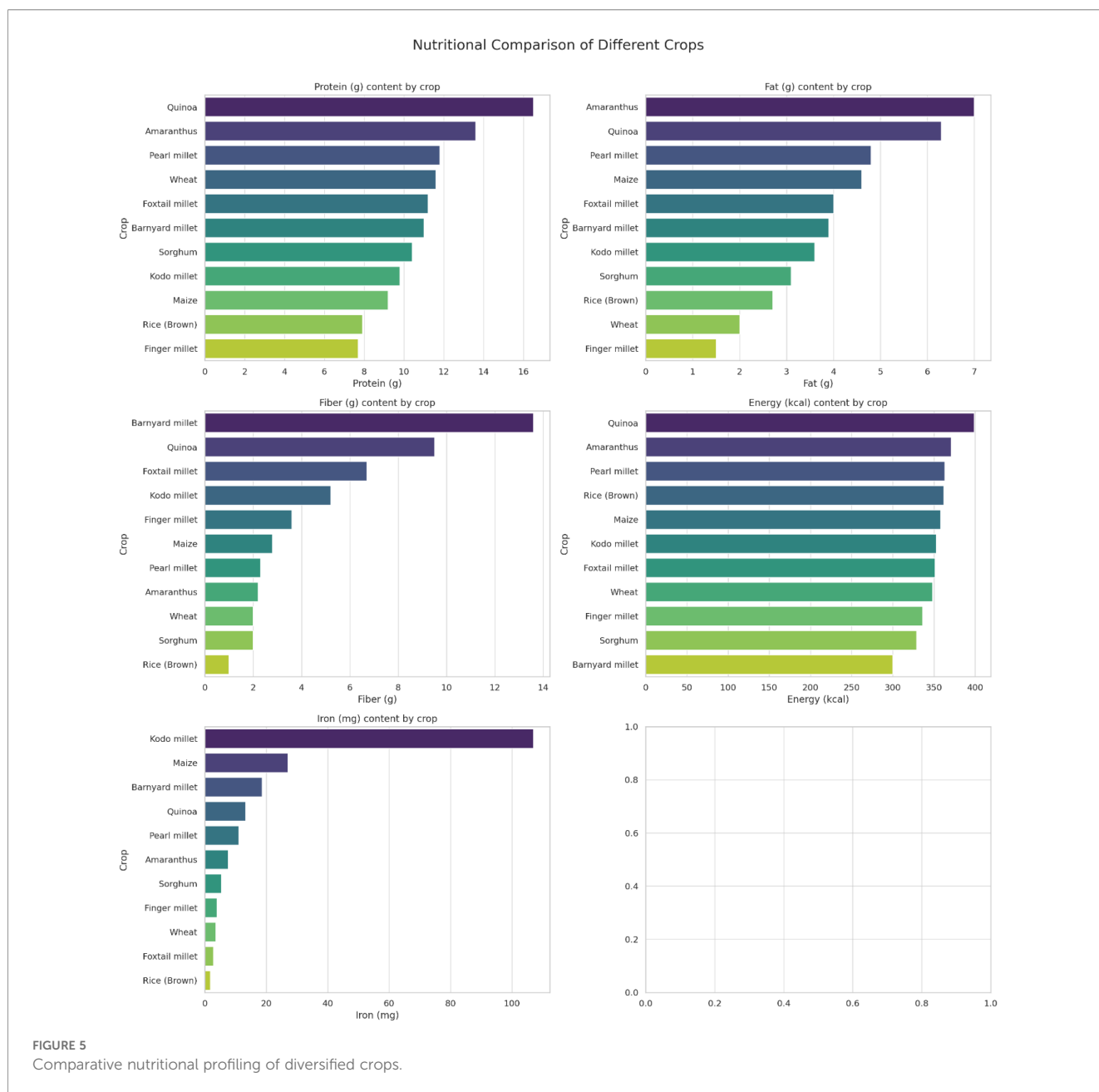
Type of crop diversification	Key benefits	References
Intercropping (e.g., legumes + cereals)	Improves resource use efficiency, enhances soil fertility, reduces pest/disease incidence	Dowling et al. (2021)
Crop Rotation (e.g., cereals–legumes–oilseeds)	Breaks pest/disease cycles, improves soil structure, boosts yield stability	Beillouin et al. (2019); Crews et al. (2018)
Agroforestry (trees with annual crops)	Enhances biodiversity, promotes carbon sequestration, improves microclimate	Cardinael et al. (2022); Altieri (1999)
Relay Cropping (second crop sown before first is harvested)	Maximizes land use efficiency, ensures continuous cover, reduces erosion	Daryanto et al. (2016)
Mixed Farming (Livestock + Crops)	Diversifies income sources, improves nutrient cycling, increases resilience	Ahmed et al. (2019)
Vertical Diversification (processing or marketing added)	Adds value to production, improves farm profitability	Berzen et al. (2023)
Horizontal Diversification (adding new crops)	Reduces market risks, enhances income stability, adapts to climate change	Abbas (2022); Devi and Sharma (2022)

and highly nutritious foods (Makate et al., 2016). Several reports reveal that diversified farms have higher dietary diversity scores and better nutritional outcomes, especially when in smallholder settings (Das et al., 2024; Ndlovu et al., 2024). For example, in India and Central Asia, the incorporation of legumes, fruits and vegetables into traditional cereal-based systems has led to better nutritional outcomes than monoculture systems (Zhang et al., 2024; Das et al., 2024). Growing crops with growing patterns also helps farmers to ensure constant growth throughout the year, which helps minimise lean periods and promotes stable food sources (Devi and Sharma, 2022; Dagunga et al., 2023). In Figure 5, nutritional comparison of different crops in series of horizontal bar charts shown this set of

charts compares the nutritional content of 11 diverse crops, including underutilized species like millets, quinoa and amaranthus. Each chart focuses on five key nutritional parameters: protein, fat, fiber, energy and iron. A normalized heatmap comparing macro and micronutrient content across five crops maize, boro rice, carrot, potato and tomato is shown in Figure 6, Macro- (carbohydrates, proteins, fats) and micronutrients (zinc, iron, Vitamins A and C) across five crops: maize, boro rice, carrot, potato, and tomato. Nutrient values are scaled from 0 to 1, where 1 indicates the highest content among all crops. The color intensity reflects this scale from light yellow (low) to dark orange (high).

TABLE 2 Economic comparison of diversified vs. monoculture systems.

Aspect	Diversified systems	Monoculture systems	References
Yield Stability	More stable over time due to resilience to pests, weather, and market shifts	High yields initially but prone to fluctuation due to climate and pest pressure	Adam and Abdulai (2024); Beillouin et al. (2019)
Gross Returns	Often higher due to multiple outputs (intercrops, rotations, livestock)	High returns possible from cash crops, but dependent on input costs and market prices	Adam and Abdulai (2024); Bommarco et al. (2013)
Input Costs	Lower in the long run due to reduced pesticide and fertilizer needs	Higher due to dependence on agrochemicals and irrigation	Beillouin et al. (2019); Abbas (2022)
Labor Requirements	Higher, often involving skilled labor; creates rural employment	Lower; favors mechanization, but fewer job opportunities	Dowling et al. (2021)
Market Risk	Reduced through diversification of products	High dependency on one commodity increases vulnerability	Adam and Abdulai (2024); Abbas (2022); Davidson (2018)
Long-term Profitability	More sustainable and profitable over time due to lower risk and improved soil health	May decline due to soil degradation, pest buildup, and input dependency	Beillouin et al. (2019); Dowling et al. (2021)
Environmental Services	Provides ecosystem services (pollination, nitrogen fixation, water regulation) that contribute to economic gains	Minimal ecosystem services; may incur external costs for restoration	Bommarco et al. (2013); Cardinael et al. (2022)
Resilience to Climate Change	Greater adaptive capacity to extreme events due to crop and income diversity	High vulnerability to droughts, floods, and pests	Beillouin et al. (2019); Davidson (2018); Chaloner et al. (2021)



3.2.4 Gender empowerment and social inclusion

Women's involvement in diversified farming systems has proven to be an important factor for improved household outcomes and community resilience (Bliznashka et al., 2023; Valencia et al., 2021). Globally, women account for almost 43% of the labour force in agriculture and also produce 60-80% of food in developing countries (Young, 2023; Ndlovu et al., 2024). Recent studies have shown that in households where women are involved in decision-making, hold assets and are active in community groups, there is a stronger tendency to grow a variety of crops with greater nutrient value (Bliznashka et al., 2023). Further, women's increased responsiveness to training and capacity-building programmes economic impact of diversification on

household nutrition and food security is even more profound (Valencia et al., 2021).

Data compiled from Food and Agriculture Organization of the United Nations (2024), Chavan and Kadam (1989); Kulczyński et al. (2019).

3.2.5 Reduction of costs and resource efficiency

Diversified cropping systems significantly lower the use of external inputs through increased natural pest control, improved soil fertility and efficient water consumption (Zhang et al., 2024). Multicropping and intercropping reduce the need for fertilizers and pesticides, thus increasing the resource use efficiency and providing a foundation for sustainable long-term productivity (Lopez et al., 2024;

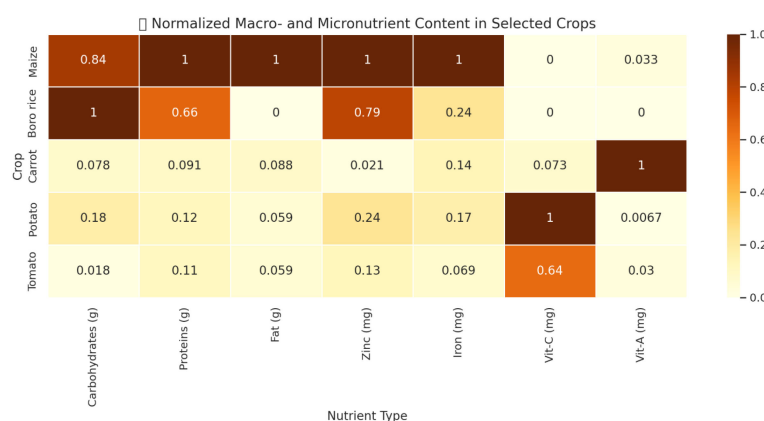


FIGURE 6

A Normalized heatmap comparing macro and micronutrient content in selected crops.

(Dagunga et al., 2023). The natural pest and disease control benefits in diversified systems significantly reduce pesticide spending. Divergent crops have different nutrient needs and nutrient uptake patterns, so by using crop rotations or planting a range of species at the same time, farmers can reduce nutrient depletion and enhance soil fertility (Devi and Sharma, 2022; Sain et al., 2023). Leguminous crops have the ability to fix atmospheric nitrogen thus enriching the soil matrix for repeated cultivations, while heterogeneous planting increases organic matter deposition resulting from various root exudates and plant residue (Sain et al., 2023; Devi and Sharma, 2022; Sain et al., 2023; Dagunga et al., 2023).

3.2.6 Poverty reduction and rural livelihood enhancement

Increased crops diversification opens several avenues for the improvement of rural livelihoods in terms of income-generating opportunities, building skills and strengthening the anthropological capacity of communities. The integration of different crops also promotes setting up new agriculturally-related industries to improve the economic capacity of the rural communities. Empirically, diversified farms (mixing livestock, poultry, horticulture, and crop production as complementary businesses) have multiple income sources that reduce the dependence on single crops (Devi and Sharma, 2022). In the Ethiopian context, a study also showed that households with diversified livelihoods were about 9% better-off in terms of poverty alleviation than those that were not diversified. Investing in non-farm livelihood activities in addition to subsistence farming leads to diversification which ensures that families become more economically better off, boost production, and better withstand environmental strains and shocks (Makate et al., 2016).

3.3 Constraints to crop diversification

3.3.1 Knowledge and skills gap

It is well known that the lack of technical knowledge is one of the main constraints to the adoption of crop diversification. A

significant percentage of farmers do not have specifically specialized knowledge that will enable them to grow different crops effectively, therefore there exists a clear-cut gap in knowledge that will restrict such diversification efforts within the farming system (Ralte and Priscilla, 2023). This problem is compounded by lack of adequate training opportunities, with most surveyed farmers complaining about unsatisfactory extension services (extension services are usually market oriented and fail to provide enough market price information, and they also do not have access to credit schemes) (Joshi and Narayan, 2019; Dagunga et al., 2023). Challenges facing NARS in providing effective training programs for diversified farming systems. There is a discernible information gap between the government services and the farmers' needs, including the inadequacy of extension services in terms of providing modern techniques and realizing participatory nature (Ralte and Priscilla, 2023). Consistently, the educational level of household heads and access to information services are identified as the key predictors of successful diversification outcomes, thus affirming the crucial importance of supporting mechanisms of sound knowledge transfer (Dalal and Shankar, 2022).

3.3.2 Access to resources

Financial constraints are significant obstacles to diversification of the crop. The lack of early start-up capital and the lack of availability of crop insurance and credit facilities have a significant impact on the ability of farmers to invest in diversification strategies. Initial investment costs for new crops, equipment and infrastructure can be prohibitive, especially for smallholder farmers who face serious capital limitations. Resource constraints go beyond monetary capital and include availability of good quality inputs as well as sufficient infrastructure and appropriate technology (Ralte and Priscilla, 2023; Ndlovu et al., 2024). Research and supply industries often focus on major commodity crops, so there may be a lack of seeds and inputs for marginal crops that could be suitable for diversification (Ralte and Priscilla, 2023). Besides, rainfed conditions, labor shortage and limited access to credit further hinder crop diversification efforts. Labor availability is another big constraint; lack of skilled labor makes it difficult for farmers to diversify. Labor scarcity, especially in peak

agricultural seasons combined with rising wages pose significant challenges to the activation of more labor resourced diversified cropping systems (Feike et al., 2012). The complexity involved in managing multiple crops at the same time adds to the need for skilled agricultural workers.

3.3.3 Policy and institutional support

Agricultural policies have tended to support monocultures and thereby created unwritten rules that make diversifying difficult. Cereal - centric policies and interventions in agri - food markets tend to diminish economic incentives for farmers to move toward diverse agri food systems (Negi et al., 2020). The absence of coordination among the public and private institutions further restricts the efficacy of the diversification strategies (Shah et al., 2021b; Ndlovu et al., 2024). Despite policy-level awareness of problems such as soil degradation and declining agrobiodiversity, implementation is fragmented and inadequate. Current approaches tend to focus on short term economic gains and ignore the long term benefits of holistic and resilient farming systems. Strengthening institutional mechanisms, and realigning policy incentives are crucial steps towards large scale diversification and sustainable agricultural transformation.

3.3.4 Market constraints

Price fluctuations and geographical isolation from markets are some of the factors that contribute to the marketing limitations in achieving successful agricultural diversification. Farmers often focus on crops that always have a high demand on the market, limiting diversification work to well-established markets for commodities that are characterised by predictable prices structures (Ralte and Priscilla, 2023; Makate et al., 2016). Value chain development of different crops is still far from developed and infrastructure support of processing, storage and transport of different crops is a far cry from major commodities.

3.3.5 Constraints of infrastructure

Infrastructure constraints represent the basic constraints on crop diversification; the lack of sufficient processing, storage and transportation facilities reduce diversification potential. The infrastructure of cultivation facilities available to farmers such as irrigation system, appropriate implement, machinery are found to be inadequate for a number of stakeholders (Ralte and Priscilla, 2023; IUCN, 2023). Limits in irrigation infrastructure are particularly constraint on water-intensive crops such as vegetables, fruits and some spices. Inadequate infrastructures of rural areas, such as road system and communication, limits connectivity between remote rural areas and urban demand centres, thereby limiting market accessibility to various diversified agricultural products (Negi et al., 2020; Makate et al., 2016). Shared-machinery services and custom harvesting operations are underdeveloped in many regions limiting the access of appropriate technology to the effective management of diverse cropping systems (Ralte and Priscilla, 2023; Makate et al., 2016).

3.3.6 Risk aversion and climate sensitivity

Farmer's risk aversion, in other words, is a major psychological and economic constraint on farmer's acceptance of crop diversification. A weak economy in terms of its capacity to carry risk presents itself as an acute constraint affecting the development of diversification (Lim, 2023). Climate change contributes to the risk perception of diversification; weather conditions that cause unpredictability in crop planning make planning for diversification of crops more complex (Ralte and Priscilla, 2023). Economic constraints combined with risk aversion lead to intertwined barriers, especially among smallholder farmers who have little financial buffer room to absorb potential losses (Lim, 2023). Insurance products based on diverse farming systems are generally lacking and farmers do not have financial protection mechanisms that can offset perceived risks associated with the introduction of new crops and management practices (Lim, 2023; Makate et al., 2016).

TABLE 3 Key case studies on crop diversification and their outcomes.

Region/ Country	Diversification strategy	Key outcomes	References
North China Plain	Diversified rotations (cash crops, legumes)	Yield increased by 38%, farm income upto 20%, reduced N ₂ O emissions by 39%, improved soil health by 45%, increased soil organic carbon by 8%	Zhang et al., 2024; Food and Agriculture Organization of the United Nations (2024)
Switzerland	Multi-sectoral working groups, value chains	Protein self-sufficiency up from 3% to 15%, improved value chain integration	DiverIMPACTS, 2022
India (Haryana)	Farm diversification, area allocation	Improved resilience, market access, higher net returns	Ralte and Priscilla, 2023
India (Odisha)	Indigenous seed revival, women-led groups	Better climate adaptation, higher yields, cultural preservation	Majhi, 2020; Paroja, 2024
Zimbabwe	Crop and livestock integration	Increased productivity, income, food security, nutrition	Makate et al., 2016
Sub-Saharan Africa	Underutilized crops (bambara, ensete, etc.)	Enhanced resilience, food security, nutrition, and cultural value	Ndlovu et al., 2024
EU (various countries)	Crop rotation, cover crops, EFAs	Biodiversity, soil health, environmental sustainability	Staniszewski and Borychowski, 2020

4 Case studies, regional insights

4.1 Global case studies

A comprehensive reports of various global case studies based on crop diversification has been presented a special importance to cropping pattern and crop rotations during adverse conditions. The most important outcomes associated with these initiatives are summarized in [Table 3](#). In addition, the evaluation of European diversification programs shows successful implementation in diverse climatic as well as administrative contexts ([Makate et al., 2016](#)). The project worked together with 25 innovation networks covering various cropping systems and diversification strategies. In Western Europe, schemes for crop diversification were grouped in 5 main clusters, which also included service crops integrating several cropping strategies involving fodder crops, cover crops and short-term leys. The analysis revealed an array of behavioural patterns and socio-economics mechanisms that are possible in terms of promoting crop diversification.

Case studies like “The Netherlands: Breaking Maize Monoculture” were mainly focused on fulfilling the societal demands for sustainable production, through a reduction in pesticide and nutrient applications. The study aimed at breaking maize monoculture and diversifying feed production. The results evidenced to the fact that fields using crop diversification performed better than the maize monoculture and thus supporting the premise that diversification improves sustainable crop production ([DiverIMPACTS, 2022](#)).

How in UK there are compelling examples of crop diversification in association with crop rotation, including cover crops and companion cropping strategies to tackle diverse agricultural challenges at the same time ([Zhang et al., 2024](#); [Makate et al., 2016](#)). Northants LEAF farmer Duncan Farrington was able to grow his cereals and oilseed rotation further to reduce blackgrass infestation, pigeon damage and disease pressures with cover crops offering useful levels of weed control while supporting crop nutrition and soil health programmes. Hungarian farms are successful in building up to an organic farm system by using 25–30% of the field for leguminous crops in rotation sequences, which brought about a rise in the biodiversity and number of pollinators, though the management complexity was a challenge for some operations. Experiments in the North China Plain give significant evidence of the benefits of biodiversity in the traditional cereal monoculture systems through cash crop and legume integration, with the diversified rotations increasing equivalent yield by as much as 38%, and reducing N₂O emissions by 39% and improving the overall greenhouse-gas balance by 88% compared with conventional wheat-maize systems. Including legumes in crop rotations stimulated microbial activity in soils, increased soil organic carbon stocks by 8% and improved the overall soil health by 45%. A large scale adoption of diversified cropping systems could potentially increase cereal production by 32% in the wheat-maize rotation with alternative crops and raise farmer’s income by 20% and realize huge environmental benefits ([Zhang et al., 2024](#)).

4.2 Regional case studies

Case studies implemented in India clarify the multifarious aspects of crop diversification in different agro-ecology zones based on information from some selected farmers in three districts of diverse environmental situations. In Haryana investigations at Kaithal (AEZ1), Hisar (AEZ2) and Bhiwani (AEZ3) explored the possibility of diversifying of farms in terms of best allocation of land, suitable number of enterprises, cost analysis of the various enterprises and net returns from diversified enterprises. These studies show the influence of local environmental conditions, market availability and farmer characteristics on diversification strategies and their final success or failure ([Ralte and Priscilla, 2023](#)). Cotton based farming systems in India are facing some specific problems and possibilities for diversification reflecting in general the constraint on the smallholder farmers. Organic cotton farms utilize a range of intercropping and crop rotation practices, but there are still substantial barriers to overcome which include the demands of the market and procurement, carrying capacities in terms of skills and plant building, supply chain, motivational issues of the farmers, and policy environment for organic cotton.

4.3 Indigenous crop revival attempts to facilitate diversification

Efforts to revive species of native crops offer significant potential for agricultural resilience and environmental sustainability and for conserving most crucial cultural heritage and traditional knowledge systems. Traditional crop varieties have a number of perspectives, such as lower input costs, easy accessibility, genetic diversity, and more resistance to climate pressures than modern varieties hybrids that most of the times require intensive external inputs. These community based initiatives focus on restoring traditional varieties that are better adapted to local environmental conditions, while maintaining genetic diversity that has been eradicated by the systematic industrial agriculture. Plant breeders are able to take advantage of the remarkable capacity of crop diversity, as it is archived in the collections of gene banks, to develop new crops and agricultural systems that can stay productive and nutritious in the face of increasing climate pressures and environmental challenges. Community based approaches have shown to be key to the success of indigenous crop revival with progressive women farmers and agricultural entrepreneurs showing leadership in the implementation of diversified crop systems that integrate traditional knowledge and appropriate modern techniques. These types of initiatives often include creating model integrated farms that combine protection structures (e.g. polyhouses, shade net houses) with farm ponds and different crop varieties which serve as biodiversity conservation centres. The sharing of traditional seeds and plants with farmers interested in cultivating heritage varieties creates networks of the exchange of knowledge and the conservation of genetic resources which have a strengthening effect on local food systems.

4.4 Insights gained/critical reflections

The creation of complex ecosystems is realized through the increase or complement of biodiversity in and outside soil, which creates favorable conditions for many beneficial organisms, such as pollinator, pest predator and soil flora and fauna. These organisms together provide crucial ecosystem services such as natural pest suppression, pollination, nutrient cycling and soil structuring. Empirical evidence shows that diversified farms can provide higher long term productivity and stability in the face of fluctuations in the environment, like droughts, pest outbreaks or the reduction in the fertility of the soil. Research shows that empowering women farmers in poor and middle income countries can create more crop diversity, which enhances the year round availability of healthy foods (Young, 2023). When women achieve increased levels of participation in decision-making for farm management, agrobiodiversity and adoption of agroecological practices is also increased. Women's involvement in agroecological social movements is positively linked to a much higher degree of empowerment in control over income and to higher levels of decision making involvement (Valencia et al., 2021). Successful diversification involves dealing with several interlinked factors at the same time such as specific crop choice, cropping for geographical and climatic conditions, characteristics of individual farmers, the development of supply chain structure and conducive institutional environments. Barriers to diversification are interlinked and exist at multiple points along supply chains and require coordinated approaches to address technical, socio-economic and policy constraints concurrently (Shah et al., 2021b).

4.5 Recommendations for policy and practice

Prioritization of cropping diversity, its ecological and resilience services need to be included in government policy and practical application to establish crop diversification as a vital component of sustainable agriculture and climate change adaptation policies. Crop diversification is a critical factor in promoting soil health, increasing biodiversity, improving livelihoods and strengthening resilience to climatic shocks, especially among smallholder farmers who are overrepresented in categories of people economically and environmentally vulnerable.

Governments should establish well-targeted incentive programmes and fiscal subsidies on the development and introduction of legume, agroforestry and other diversifying systems appropriate to ecological conditions in the region: At the same time, significant investments are needed in farmer capacity building, market linkage development and research activities to develop scalable, context-sensitive models. Moreover, diversification training modules, curricula and inclusion of extension services will raise awareness and create wider adoption.

On the farm level, diversification provides many practical benefits, such as risk-mitigation, a mitigation of agrochemical

dependency, increased soil productivity and synergy with other practices related to climate-smart and regenerative agriculture. Secondly, to ensure long term and effective adoption it is crucial that policy frameworks are participative and inclusive and that insights from farmers, researchers and community stakeholders are taken into account to craft interventions that are locally appropriate.

5 Recommendations

A critical appraisal of the insufficient adoption of diversification strategies at the grass root level in crop diversification is required for clarifying the source of problems experienced by farmers. This assessment helps to form targeted interventions to create awareness campaigns, commercial schemes and other enabling structures to bring out the most benefits.

A tailored enhancement of extension service and farmer training programs to reduce potential knowledge gaps and build capacity for successful diversification practice adoption is a major requirement. Farmers need a strong information dissemination system to make rational decisions on their combinations of crops and the establishments of pest-disease management strategy. Furthermore, participatory research and farmer-to-farmer knowledge transfers allow diversification measures to be adapted to the local contingencies and to ensure that innovations are suitable to their intended targets.

Policy formulation and reform that promote women's participation, increase infrastructure, improve access to markets and boost research to support farmers in making the switch to diversified cropping systems is posited. Consequently, both research and documentation help to build the knowledge base in the long term while supporting economic and environmental benefits and the development of context-specific measures. Studies on using indigenous knowledge and traditional varieties of crops adapted well to specific agro-climatic and soil conditions are necessary to increase sustainability and resilience.

6 Future prospects

Despite the strong body of evidence in favour of the concept of crop diversification, the present review highlights certain gaps related to research, extension, policy formulation and inclusive decision-making between grassroots farmers, researchers and related stakeholders. The larger applications of crop diversification are often spatially localized and carried out under controlled conditions and may limit the generality of results and the visibility of some of the practical problems faced by farmers. Discrepancies in research design and methodology hinder cross-comparison and documentation of findings, further complicating the scale up of such interventions. India's various agro-climatic zones, having their own unique soil types, climatic conditions, water availability and socio-economic situation of farmers, favour specific

crop or crop combination over the others. Accounting for this diversity at all levels while developing the research agendas is essential for achieving the research goals. Economic aspects (such as cost-efficiency, labour requirements and viability of diversification strategies in the market) is not sufficiently disclosed, making it difficult to judge the complete economic feasibility of such diversification strategies under different agro-climatic conditions. Moreover, long-term viability of diversified systems in different climatic regimes is under-researched and interaction of diversification and other agricultural practices (e.g. irrigation, mechanization, and pests management) as well as the adoptability of emerging technologies (e.g., remote sensing and artificial intelligence used for monitoring and improving diversification) are yet to be well-researched representing notable research gaps for future studies.

7 Conclusion

Crop diversification is a proven strategy for helping to increase the resilience, productivity and sustainability of the world's agricultural systems. It stabilises farm incomes by spreading risk related to multiple crops and avenues of markets as this reduces vulnerability to climatic and market shocks. Diversification also enhances food security by enhancing the abundance and accessibility of diversity and nutrition of food and especially in smallholder and resource poor households where environmental benefits are just as important. Diversified systems are part of the solution for soil health, biodiversity and climate change by increasing carbon sequestration and decreasing greenhouse gas emissions. The use of legume cover crops and agroforestry further enhances these ecological benefits. However, successful adoption of diversified systems is hampered by multiple barriers such as availability of knowledge and skills, availability of quality seed and inputs, quality of extension services provided and poor infrastructure in rural areas with low production potential in rural areas restricting farmers from implementing diversification. Policy distortions, e.g. subsidies for staple crops and limited market access for non-traditional crops, are additional disincentives for diversification. Risk aversion, especially by smallholders, is also delaying larger transition to more resilient and sustainable farming systems. In order to fully exploit the potential of crop diversification, concerted efforts from governments, researchers, extension services and farming communities are needed. By tackling the identified barriers as well as leveraging the best available research and technology, diversification can be taken at scale and work to create resilient, productive and sustainable agricultural landscapes around the world.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material. Further inquiries can be directed to the corresponding author.

Author contributions

RS: Supervision, Formal analysis, Writing – original draft, Data curation, Software, Methodology, Funding acquisition, Conceptualization, Resources, Validation, Visualization, Investigation, Project administration, Writing – review & editing. LL: Validation, Writing – review & editing, Supervision. AP: Conceptualization, Writing – review & editing, Visualization. PB: Investigation, Writing – review & editing, Supervision. HJ: Funding acquisition, Validation, Investigation, Supervision, Writing – review & editing. DH: Writing – original draft, Conceptualization, Data curation. KV: Writing – review & editing, Methodology, Conceptualization. AS: Supervision, Validation, Writing – review & editing. KS: Writing – original draft, Data curation. KA: Data curation, Methodology, Writing – review & editing. MP: Writing – original draft, Resources, Methodology.

Funding

The author(s) declared that financial support was not received for this work and/or its publication.

Acknowledgments

All the authors gratefully acknowledge their parent institute for its unwavering support and infrastructural assistance in carrying out this research.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was used in the creation of this manuscript.

Any alternative text (alt text) provided alongside figures in this article has been generated by Frontiers with the support of artificial intelligence and reasonable efforts have been made to ensure accuracy, including review by the authors wherever possible. If you identify any issues, please contact us.

Publisher's note

All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.

References

- Aare, A. K., Egmose, J., Lund, S., and Hauggaard-Nielsen, H. (2021). Opportunities and barriers in diversified farming and the use of agroecological principles in the Global North—The experiences of Danish biodynamic farmers. *Agroecol Sustain Food Syst.* 45, 390–416. doi: 10.1080/21683565.2020.1822980
- Abbas, S. (2022). Climate change and major crop production: evidence from Pakistan. *Environ. Sci. Pollut. Res.* 29, 5406–5414. doi: 10.1007/s11356-021-16041-4
- Adam, B., and Abdulai, A. (2024). Heterogeneous impact of crop diversification on farm net returns and risk exposure: Empirical evidence from Ghana. *Can. J. Agric. Econ.* 72 (4), 469–487. doi: 10.1111/cjag.12360
- Ahmed, I., Ullah, A., Rahman, M. H., Ahmad, B., Wajid, S. A., Ahmad, A., et al. (2019). “Climate change impacts and adaptation strategies for agronomic crops,” in *Climate change and agriculture* (London, United Kingdom: IntechOpen). doi: 10.5772/INTECHOPEN.82697
- Akther, S., and Evans, J. (2024). Emerging attributes of adaptive governance in the global south. *Front. Environ. Sci.* 12, 1372157. doi: 10.3389/fenvs.2024.1372157
- Almagro, M., Re, P., Diaz-Pereira, E., Boix-Fayos, C., Sánchez-Navarro, V., Zornoza, R., et al. (2023). Crop diversification effects on soil organic carbon and nitrogen storage and stabilization is mediated by soil management practices in semiarid woody crops. *Soil Tillage Res.* 233, 105815. doi: 10.1016/j.still.2023.105815
- Altieri, M. A. (1999). The ecological role of biodiversity in agroecosystems. *Agric. Ecosyst. Environ.* 74, 19–31. doi: 10.1016/S0167-8809(99)00028-6
- Azani, N., Ghaffar, M. A., Suhaimi, H., Azra, M. N., Hassan, M. M., Jung, L. H., et al. (2021). The impacts of climate change on plankton as live food: a review. *IOP Conf Ser. Earth Environ. Sci.* 869, 012005. doi: 10.1088/1755-1315/869/1/012005
- Baffes, J., and Nagle, P. (2022). *Commodity markets: evolution, challenges, and policies* (Washington, DC: World Bank Publications).
- Beillouin, D., Ben-Ari, T., and Makowski, D. (2019). Evidence map of crop diversification strategies at the global scale. *Environ. Res. Lett.* 14, 123001. doi: 10.1088/1748-9326/ab4449
- Beillouin, D., Ben-Ari, T., Malézieux, E., Seufert, V., and Makowski, D. (2021). Positive but variable effects of crop diversification on biodiversity and ecosystem services. *Glob Chang Biol.* 27, 4697–4710. doi: 10.1111/GCB.15747
- Bernzen, A., Sohns, F., Jia, Y., and Braun, B. (2023). Crop diversification as a household livelihood strategy under environmental stress. *Land Use Policy* 132, 106796. doi: 10.1016/j.landusepol.2023.106796
- Biber-Freudenberger, L., Ziemacki, J., Tonnang, H. E., and Borgemeister, C. (2016). Future risks of pest species under changing climatic conditions. *PLoS One* 11, e0153237. doi: 10.1371/journal.pone.0153237
- Bliznashka, L., Gillespie, S., and van den Bold, M. (2023). Women’s empowerment and crop diversity in low- and middle-income countries: A systematic review. *Glob Food Sec* 38, 100670. doi: 10.1016/j.gfs.2023.100670
- Bogado, A. C. S., Estrada-Carmona, N., Beillouin, D., Chéron-Bessou, C., Rapidel, B., and Jones, S. K. (2024). Farming for the future: understanding factors enabling the adoption of diversified farming systems. *Glob Food Sec* 43, 100820. doi: 10.1016/j.gfs.2024.100820
- Bommarco, R., Kleijn, D., and Potts, S. G. (2013). Ecological intensification: harnessing ecosystem services for food security. *Trends Ecol. Evol.* 28, 230–238. doi: 10.1016/j.tree.2012.10.012
- Bradshaw, C., Eyre, D., Korycinska, A., Li, C., Steynor, A., and Kriticos, D. (2024). Climate change in pest risk assessment: interpretation and communication of uncertainties. *EPPO Bull.* 54, 4–19. doi: 10.1111/epp.12985
- Briones, R. M., and Rakotoarisoa, M. A. (2013). *Investigating the structures of agricultural trade industry in developing countries* (Rome, Italy: FAO Commodity and Trade Policy Research Working Paper), 38.
- Bybee-Finley, K. A., and Ryan, M. R. (2018). Advancing intercropping research and practices in industrialized agricultural landscapes. *Agriculture* 8, 80. doi: 10.3390/agriculture8060080
- Cardinael, R., Umulisa, V., and Tully, K. (2022). Agroforestry and carbon sequestration: A global synthesis. *Nat. Sustain* 5, 512–522.
- Chaloner, T., Gurr, S., and Bebbler, D. (2021). Plant pathogen infection risk tracks global crop yields under climate change. *Nat. Clim Change* 11, 710–715. doi: 10.1038/s41558-021-01104-8
- Chamkhi, I., Cheto, S., Geistlinger, J., Zeroual, Y., Kouisni, L., Bargaz, A., et al. (2022). Legume-based intercropping systems promote beneficial rhizobacterial community and crop yield under stressing conditions. *Ind. Crops Prod* 183, 114958. doi: 10.1016/j.indcrop.2022.114958
- Chapman, S., Birch, C. E., Pope, E., Sallu, S., Bradshaw, C., Davie, J., et al. (2020). Impact of climate change on crop suitability in sub-Saharan Africa in parameterized and convection-permitting regional climate models. *Environ. Res. Lett.* 15 (9), 094086. doi: 10.1088/1748-9326/ab9daf
- Chavan, J. K., and Kadam, S. S. (1989). Nutritional improvement of cereals by fermentation. *Crit. Rev. Food Sci. Nutr.* 28, 349–400. doi: 10.1080/10408398909527507
- Chellam, S., Bai, D., Vijaya Rani, D., Sindhu, M., Pushpalatha, V., JS, R., et al. (2024). Agro-ecological approaches to pest management: The role of crop diversification and habitat manipulation. *Int. J. Adv. Biochem. Res.* 8, 154–157. doi: 10.33545/26174693.2024.v8.i9sb.2077
- Clements, R., Haggard, J., Quezada, A., and Torres, J. (2011). *Technologies for Climate Change Adaptation – Agriculture Sector*. Ed. X. Zhu (Roskilde: UNEP Risø Centre). Available online at: <http://tech-action.org/> (Accessed March 30, 2025).
- Cozím-Melges, F., Ripoll-Bosch, R., Veen, G. F., Oggiano, P., Bianchi, F. J., van der Putten, W. H., et al. (2024). Farming practices to enhance biodiversity across biomes: a systematic review. *NPJ Biodivers* 3, 1. doi: 10.1038/s44185-023-00034-2
- Crews, T. E., Carton, W., and Olsson, L. (2018). Is the future of agriculture perennial? Imperatives and opportunities to reinvent agriculture by shifting from annual monocultures to perennial polycultures. *Glob Sustain* 1, e11. doi: 10.1017/sus.2018.11
- Dagunga, G., Ayamga, M., Laube, W., Ansah, I. G. K., Kornher, L., and Kotu, B. H. (2023). Agroecology and resilience of smallholder food security: A systematic review. *Front. Sustain Food Syst.* 7. doi: 10.3389/fsufs.2023.1267630
- Dalal, S., and Shankar, R. (2022). Constraints and strategies for crop diversification in India. *Indian J. Ext Educ.* 58, 1–10.
- Daryanto, S., Wang, L., and Jacinthe, P. A. (2016). Global synthesis of drought effects on cereal, legume, tuber and root crops production: A review. *Agric. Water Manag* 179, 18–33. doi: 10.1016/j.agwat.2016.04.022
- Das, A., Ramesh, P., Babu, S., and Singh, R. (2024). Challenges for crop diversification in cotton-based farming systems in India. *Front. Agron.* 6. doi: 10.3389/fagro.2024.1370878
- Dasgupta, K., and Mahanty, A. (2024). Carbon sequestration in a changing climate: Management techniques and strategic solutions. *Asian Res. J. Agric.* 17, 703–713. doi: 10.9734/arja/2024/v17i4577
- Davidson, D. J. (2018). Rethinking adaptation: emotions, evolution, and climate change. *Nat. Cult* 13, 378–402. doi: 10.3167/nc.2018.130304
- Devi, S., and Sharma, R. (2022). Crop diversification and its impact on farm income and employment: Evidence from India. *Int. J. Agric. Food Sci.* 7, 13–18. Available online at: <https://www.agriculturaljournals.com/archives/2025/vol7issue1/PartA/7-1-4-787.pdf> (Accessed September 20, 2025).
- DiverIMPACTS (2022). *Diversified cropping systems: European innovation projects*. Available online at: <https://diverimpacts.net> (Accessed April 10, 2025).
- Dowling, A., Sadras, V. O., Roberts, P., Doolette, A., Zhou, Y., and Denton, M. D. (2021). Legume-oilseed intercropping in mechanised broadacre agriculture—a review. *Field Crop Res.* 260, 107980. doi: 10.1016/j.fcr.2020.107980
- Drinkwater, L. E., Wagoner, P., and Sarantonio, M. (1998). Legume-based cropping systems have reduced carbon and nitrogen losses. *Nature* 396, 262–265. doi: 10.1038/24376
- Elahi, E., Khalid, Z., Tauni, M. Z., Zhang, H., and Lirong, X. (2022). Extreme weather events risk to crop production and the adaptation of innovative management strategies to mitigate the risk: a retrospective survey of rural Punjab, Pakistan. *Technovation* 117, 102255. doi: 10.1016/j.technovation.2021.102255
- Ericksen, P. J., Ingram, J. S., and Liverman, D. M. (2009). Food security and global environmental change: emerging challenges. *Environ. Sci. Policy* 12, 373–377. doi: 10.1016/j.envsci.2009.04.007
- Food and Agriculture Organization of the United Nations (2024). *The state of the world’s land and water resources for food and agriculture – Systems at breaking point: Synthesis report 2021*. (Rome, Italy: FAO). doi: 10.4060/cb7654en
- Feike, T., Chen, Q., Graeff-Hönniger, S., Pfenning, J., and Claupein, W. (2012). The economic performance of crop production in the North China Plain: A case study of maize, wheat, and cotton. *Agric. Syst.* 109, 1–9. doi: 10.1016/j.jagsy.2012.01.002
- Gawdiya, S., Sharma, R. K., Singh, H., and Kumar, D. (2025). Crop diversification as a cornerstone for sustainable agroecosystems: tackling biodiversity loss and global food system challenges. *Discov. Appl. Sci.* 7, 1–16. doi: 10.1007/s42452-025-06855-z
- Gomez-Zavaglia, A., Mejuto, J. C., and Simal-Gandara, J. (2020). Mitigation of emerging implications of climate change on food production systems. *Food Res. Int.* 134, 109256. doi: 10.1016/j.foodres.2020.109256
- Grahmann, K., Reckling, M., Hernández-Ochoa, I., Donat, M., Bellingrath-Kimura, S., and Ewert, F. (2024). Co-designing a landscape experiment to investigate diversified cropping systems. *Agric. Syst.* 217, 103950. doi: 10.1016/j.jagsy.2024.103950
- Hassan, M. A., Xiang, C., Farooq, M., Muhammad, N., Yan, Z., Hui, X., et al. (2021). Cold stress in wheat: plant acclimation responses and management strategies. *Front. Plant Sci.* 12. doi: 10.3389/fpls.2021.676884
- Hien, N. T. (2025). Crop diversification and sustainable agricultural development: Evidence from Vietnam. *J. Agribus Rural Dev.* 17, 1–12.
- IUCN (2023). Crop diversification practice. *Int. Union Conserv. Nat.* Available online at: <https://www.iucn.org/resources/issues-brief/crop-diversification>.

- Jaworski, C. C., Thomine, E., Rusch, A., Lavoie, A.-V., Wang, S., and Desneux, N. (2023). Crop diversification to promote arthropod pest management: A review. *Agric. Commun.* 1 (1), 100004. doi: 10.1016/j.agrcom.2023.100004
- Jiang, H., Du, Y., Han, W., Wang, L., Xiang, C., Ge, Y., et al. (2022). Assembling plant diversity mitigates greenhouse gas emissions and achieves high nitrogen removal when treating the low-C/N wastewater by constructed wetlands. *Environ. Sci. Pollut. Res.* 30, 228–241. doi: 10.1007/s11356-022-22088-8
- Joshi, P. K., and Narayan, R. (2019). Crop diversification in India: Trends, determinants, and policy implications. *Indian J. Agric. Econ.* 74, 299–313. Available online at: <https://www.isaeindia.org/wp-content/uploads/2019/12/4.pdf> (Accessed April 10, 2025).
- Kabange, N. R., Kwon, Y., Lee, S.-M., Kang, J.-W., Cha, J.-K., Park, H., et al. (2023). Mitigating greenhouse gas emissions from crop production and management practices, and livestock: A review. *Sustainability* 15 (22), 15889. doi: 10.3390/su152215889
- Kanter, D. R., Musumba, M., Wood, S., Palm, C., and McLaren, J. (2021). A framework for assessing the sustainability of nutrient management. *Nat. Food* 2 (2), 91–97. doi: 10.1016/j.nfs.2016.09.010
- Kastner, T., Chaudhary, A., Gingrich, S., Marques, A., Persson, U. M., Bidoglio, G., et al. (2021). Global agricultural trade and land system sustainability: implications for ecosystem carbon storage, biodiversity, and human nutrition. *One Earth* 4 (10), 1425–1443. doi: 10.1016/j.oneear.2021.09.006
- Koontz, T. M., Gupta, D., Mudliar, P., and Ranjan, P. (2015). Adaptive institutions in social-ecological systems governance: a synthesis framework. *Environ. Sci. Policy* 53, 139–151. doi: 10.1016/j.envsci.2015.01.003
- Kulczyński, B., Kobus-Cisowska, J., Taczanowski, M., Kmiecik, D., and Gramza-Michałowska, A. (2019). The chemical composition and nutritional value of chia seeds—current state. *Nutrients* 11 (6), 1242. doi: 10.3390/nu11061242
- Kumar, S., and Gupta, S. (2015). Crop diversification towards high-value crops in India: A state level empirical analysis. *Agric. Econ. Res. Rev.* 28, 339–350. doi: 10.5958/0974-0279.2016.00012.4
- Lakhran, H., Kumar, S., and Bajji, R. (2017). Crop diversification: an option for climate change resilience. *Trends Biosci.* 10, 516–518.
- Lal, R. (2004). Soil carbon sequestration to mitigate climate change. *Geoderma* 123, 1–22. doi: 10.1016/j.geoderma.2004.01.032
- Lenné, J., and Wood, D. (2024). Crop diversity in agroecosystems for pest management and food production. *Plants* 13(8), 1164. doi: 10.3390/plants13081164
- Letourneau, D. K., Armbricht, I., Rivera, B. S., Lerma, J. M., Carmona, E. J., Daza, M. C., et al. (2011). Does plant diversity benefit agroecosystems? A synthetic review. *Ecol. Appl.* 21 (1), 9–21. doi: 10.1890/09-2026.1
- Li, G., Yu, C., Shen, P., Hou, Y., Ren, Z., Li, N., et al. (2023). Crop diversification promotes soil aggregation and carbon accumulation in global agroecosystems: a meta-analysis. *J. Environ. Manage.* 350, 119661. doi: 10.1016/j.jenvman.2023.119661
- Lim, S. S. (2023). Risk aversion, crop diversity, and food security: Evidence from rural Ethiopia. *Food Policy* 117, 102551. doi: 10.1016/j.foodpol.2023.102551
- Lipper, L., Thornton, P., Campbell, B. M., Baedeker, T., Braimah, A., Bwalya, M., et al. (2014). Climate-smart agriculture for food security. *Nat. Clim. Change* 4 (12), 1068–1072. doi: 10.1038/nclimate2437
- Lóczy, D., Dezső, J., Weidinger, T., Horváth, L., Pirkhoffer, E., and Czigány, S. (2024). Soil moisture conservation through crop diversification and related ecosystem services in a blown-sand area with high drought hazard. *Plants* 13 (4), 494. doi: 10.3390/plants13040494
- Lopez, B., Haddad, N. M., Tilman, D., and MacDougall, A. S. (2024). Joint environmental and social benefits from diversified agriculture. *Science* 384, 60–65. doi: 10.1126/science.adj1914
- Kishore, S. M., Renukaswamy, N. S., and Abhishek, V. (2024). Addressing climate change: The role of agriculture in greenhouse gas mitigation. *Asian Res. J. Agric.* 17 (4), 731–739. doi: 10.9734/arja/2024/v17i4581
- Maitra, S., Hossain, A., Brestic, M., Skalicky, M., Ondrisik, P., Gitari, H., et al. (2021). Intercropping—a low input agricultural strategy for food and environmental security. *Agronomy* 11 (2), 343. doi: 10.3390/agronomy11020343
- Majhi, K. (2020). *Return of indigenous crops helps reduce farm distress and restore ecosystems* (Mongabay India). Available online at: <https://india.mongabay.com/2020/08/return-of-indigenous-crops-helps-reduce-farm-distress-and-restore-ecosystems/>.
- Makate, C., Wang, R., Makate, M., and Mango, N. (2016). Crop diversification and livelihoods of smallholder farmers in Zimbabwe: Adaptive management for environmental change. *PLoS One* 11, e0159845. doi: 10.1371/journal.pone.0159845
- Malhi, G. S., Kaur, M., and Kaushik, P. (2021). Impact of climate change on agriculture and its mitigation strategies: a review. *Sustainability* 13, 1318. doi: 10.3390/su13031318
- McCauley, K., and Barlow, K. (2023). Regenerative agriculture: increasing plant diversity and soil carbon sequestration on agricultural landscapes. *Surg Journal*, 15 (1). doi: 10.21083/surg.v15i1.7196
- McCord, P. F., Cox, M., Schmitt-Harsh, M., and Evans, T. (2015). Crop diversification as a smallholder livelihood strategy within semi-arid agricultural systems near Mount Kenya. *Land Use Policy* 42, 738–750. doi: 10.1016/j.landusepol.2014.10.012
- Misra, A. K. (2014). Climate change and challenges of water and food security. *Int. J. Sustain. Built Environ.* 3, 153–165. doi: 10.1016/j.ijsbe.2014.04.006
- Molénat, J., Barkaoui, K., Benyoussef, S., Mekki, I., Zitouna, R., and Jacob, F. (2023). Diversification from field to landscape to adapt Mediterranean rainfed agriculture to water scarcity in climate change context. *Curr. Opin. Environ. Sustain.* 65, 101336. doi: 10.1016/j.cosust.2023.101336
- Mortensen, D. A., and Smith, R. G. (2020). Confronting barriers to cropping system diversification. *Front. Sustain. Food Syst.* 4, 564197. doi: 10.3389/fsufs.2020.564197
- Muhie, S. H. (2022). Novel approaches and practices to sustainable agriculture. *J. Agric. Food Res.* 10, 100446. doi: 10.1016/j.jafr.2022.100446
- Nandi, R., Krupnik, T. J., and Kabir, W. (2024). Crop diversification in Bangladesh: public policy provisions, practices, and insights for future initiatives. *J. Agric. Food Res.* 18, 101486. doi: 10.1016/j.jafr.2024.101486
- Nasiro, K. (2024). Cropping systems diversification as an approach to enhancing crop productivity: a review. *Plant* 12 (3), 48–65. doi: 10.11648/j.plant.20241203.12
- Nawaz, T., Gu, L., Fahad, S., Saud, S., Harrison, M. T., and Zhou, R. (2024a). Sustainable protein production through genetic engineering of cyanobacteria and use of atmospheric N₂ gas. *Food Energy Secur* 13, e536. doi: 10.1002/fes3.536
- Nawaz, T., Saud, S., Gu, L., Khan, I., Fahad, S., and Zhou, R. (2024b). Cyanobacteria: harnessing the power of microorganisms for plant growth promotion, stress alleviation, and phytoremediation in the era of sustainable agriculture. *Plant Stress* 11, 100399. doi: 10.1016/j.stress.2024.100399
- Ndlovu, M., Scheelbeek, P., Ngidi, M., and Mabhaudhi, T. (2024). Underutilized crops for diverse, resilient and healthy agri-food systems: a systematic review of sub-Saharan Africa. *Front. Sustain. Food Syst.* 8. doi: 10.3389/fsufs.2024.1498402
- Negi, D. S., Bithal, P. S., Roy, D., and Hazrana, J. (2020). Market access, price policy and diversification in Indian agriculture. *Agric. Econ. Res. Rev.* 33, 1–12.
- Paroja, S. (2024). *How indigenous women farmers are reviving lost crops in Odisha* (Mongabay India). Available online at: <https://india.mongabay.com/2024/02/how-indigenous-women-farmers-are-reviving-lost-crops-in-odisha/>.
- Peralta, A. L., Sun, Y., Sun, Y., McDaniel, M. D., and Lennon, J. T. (2018). Crop rotational diversity increases disease suppressive capacity of soil microbiomes. *Ecosphere* 9, e02235. doi: 10.1002/ECS2.2235
- Pramanick, B., Kumar, M. R., Naik, B. M., Singh, S. K., Kumar, M., and Singh, S. V. (2023). Soil carbon nutrient cycling, energetics, and carbon footprint in calcareous soils with adoption of long term conservation tillage practices and cropping systems diversification. *Sci. Total Environ.* 912, 169421. doi: 10.1016/j.scitotenv.2023.169421
- Pretty, J. (2011). Interdisciplinary progress in approaches to address social ecological and ecocultural systems. *Environ. Conserv.* 38, 127–139. doi: 10.1017/S0376892910000937
- Pretty, J., and Bharucha, Z. P. (2014). Sustainable intensification in agricultural systems. *Ann. Bot.* 114, 1571–1596. doi: 10.1093/aob/mcu205
- Ralte, R., and Priscilla, L. (2023). *Crop Diversification in India: A Review*. 135–142. Available online at: <https://renupublishers.com/images/article/IJB5V10n1n.pdf> (Accessed June 04, 2025).
- Roysen, R., Bruehwiler, N., Kos, L., Boyer, R., and Koehrsen, J. (2024). Rethinking the diffusion of grassroots innovations: an embedding framework. *Technol. Forecast. Soc. Chang* 200, 123156. doi: 10.1016/j.techfore.2023.123156
- Sain, G., Magrini, E., Murgue, C., and Thomas, M. (2023). Overcoming barriers to crop diversification uptake in Europe: a literature review. *Front. Sustain. Food Syst.* 7. doi: 10.3389/fsufs.2023.1107700
- Saleem, A., Anwar, S., Nawaz, T., Fahad, S., Saud, S., Rahman, T., et al. (2024). Securing a sustainable future: the climate change threat to agriculture, food security, and sustainable development goals. *J. Umm Al Qura Univ Appl. Sci.* 11 (3), 595–611. doi: 10.1007/s43994-024-00177-3
- Shah, K. K., Modi, B., Pandey, H. P., Subedi, A., Aryal, G., Pandey, M., et al. (2021a). Diversified crop rotation: an approach for sustainable agriculture production. *Adv. Agric.* 2021, 8924087. doi: 10.1155/2021/8924087
- Shah, S., Singh, R., and Singh, V. (2021b). Selection of incentives for a business strategy based on crop diversification. *Span J. Agric. Res.* 22, e0104. doi: 10.5424/sjar/2024223-19967
- Shammin, M. R., Haque, A. E., and Faisal, I. M. (2022). “A framework for climate resilient community based adaptation,” in *Climate Change and Community Resilience*. Eds. K. E. Haque, P. Mukhopadhyay, M. Nepal and M. R. Shammin (Springer, Singapore), 11–30.
- Shrestha, S. (2019). Effects of climate change in agricultural insect pest. *Acta Sci. Agric.* 3, 74–80. doi: 10.31080/ASAG.2019.03.0727
- Smith, P., Martino, D., Cai, Z., Gwary, D., Janzen, H., Kumar, P., et al. (2020). Greenhouse gas mitigation in agriculture. *Nat. Food* 363 (1492), 789–813. doi: 10.1098/rstb.2007.2184
- Song, X., Wang, X., Li, X., Zhang, W., and Scheffran, J. (2021). Policy oriented versus market induced: factors influencing crop diversity across China. *Ecol. Econ* 190, 107184. doi: 10.1016/j.ecolecon.2021.107184
- Souissi, A., Dhehibi, B., Oumer, A. M., Mejri, R., Fria, A., Zlaoui, M., et al. (2024). Linking farmers’ perceptions and management decision toward sustainable agroecological transition: evidence from rural Tunisia. *Front. Nutr.* 11, 1389007. doi: 10.3389/fnut.2024.1389007

- Sridhar, R., Pilla, A., Bharteey, P. K., Jatav, H. S., Longkumer, L. T., Singh, A. P., et al. (2025). Integrating emerging technologies and eco-friendly materials for soil health and environmental resilience. *Research in Ecology*, 7 (5), 179–204. doi: 10.30564/re.v7i5.11439
- Staniszewski, J., and Borychowski, M. (2020). Relationship between crop diversification and farm efficiency. *Eur. Rev. Agric. Econ* 47, 1612–1646. doi: 10.1093/erae/jbaa016
- Sturrock, R. N., Frankel, S. J., Brown, A. V., Hennon, P. E., Kliejunas, J. T., Lewis, K. J., et al. (2011). Climate change and forest diseases. *Plant Pathol.* 60, 133–149. doi: 10.1111/j.1365-3059.2010.02406.x
- Szyniszewska, A. M., Akrivou, A., Björklund, N., Boberg, J., Bradshaw, C., Damus, M., et al. (2024). Beyond the present: how climate change is relevant to pest risk analysis. *EPPO Bull.* 54, 20–37. doi: 10.1111/epp.12986
- Tamburini, G., Bommarco, R., Wanger, T. C., Kremen, C., van der Heijden, M. G., Liebman, M., et al. (2020). Agricultural diversification promotes multiple ecosystem services without compromising yield. *Sci. Adv.* 6, eaba1715. doi: 10.1126/sciadv.aba1715
- Temesgen, H., Wu, W., Legesse, A., and Yirsaw, E. (2021). Modeling and prediction of effects of land use change in an agroforestry dominated southeastern Rift Valley escarpment of Ethiopia. *Remote Sens Appl. Soc. Environ.* 21, 100469. doi: 10.1016/j.rsase.2021.100469
- Tilman, D., Cassman, K. G., Matson, P. A., Naylor, R., and Polasky, S. (2002). Agricultural sustainability and intensive production practices. *Nature* 418, 671–677. doi: 10.1038/nature01014
- Valencia, V., Wittman, H., and Blesh, J. (2021). Women's empowerment, production choices, and crop diversity in low- and middle-income countries. *Nat. Food* 2, 681–688. doi: 10.1038/s43016-021-00355-4
- Vernooy, R. (2022). Does crop diversification lead to climate related resilience? Improving the theory through insights on practice. *Agroecol Sustain Food Syst.* 46, 877–901. doi: 10.1080/21683565.2022.2076184
- Wang, B., Wang, G., van Dam, J., Yang, X., Ritsema, C., Siddique, K. H. M., et al. (2024). Diversified crop rotations improve crop water use and subsequent cereal crop yield through soil moisture compensation. *Agric. Water Manag.* 294, 108721. doi: 10.1016/j.agwat.2024.108721
- Whitton, J., and Carmichael, A. (2024). Systemic barriers preventing farmer engagement in the agricultural climate transition: a qualitative study. *Sustain Sci.* doi: 10.1007/s11625-024-01504-7
- Yan, Z., Zhou, J., Liu, C., Jia, R., Mganga, K. Z., Yang, L., et al. (2023). Legume based crop diversification reinforces soil health and carbon storage driven by microbial biomass and aggregates. *Soil Tillage Res.* 234, 105848. doi: 10.1016/j.still.2023.105848
- Yang, X., Xiong, J., Du, T., Ju, X., Gan, Y., Liu, S., et al. (2024). Diversifying crop rotation increases food production, reduces net greenhouse gas emissions and improves soil health. *Nat. Commun.* 15 (1), 198. doi: 10.1038/s41467-023-44464-9
- You, Y., Ting, M., and Biasutti, M. (2024). Climate warming contributes to the record shattering 2022 Pakistan rainfall. *NPJ Clim Atmos Sci.* 7, 1. doi: 10.1038/s41612-024-00630-4
- Young, S. L. (2023). Empowering women in agriculture: pathways to food security and crop diversity. *Glob Food Sec* 38, 100670. doi: 10.1016/j.gfs.2023.100670
- Zhang, Y., Wang, Y., Liu, X., Zhang, W., and Zuo, L. (2024). Diversifying crop rotation increases food production, reduces net greenhouse gas emissions, and improves soil health in the North China Plain. *Nat. Food* 5, 1–12. doi: 10.1038/s43016-023-00915-6
- Zou, Y., Liu, Z., Chen, Y., Wang, Y., and Feng, S. (2024). Crop rotation and diversification in China: enhancing sustainable agriculture and resilience. *Agriculture* 14, 1465. doi: 10.3390/agriculture14091465